CHAPTER 1

HVAC SYSTEM ANALYSIS AND SELECTION

AN HVAC system maintains desired environmental conditions in a space. In almost every application, many options are available to the design engineer to satisfy a client's building program and design intent. In the analysis, selection, and implementation of these options, the design engineer should consider the criteria defined here, as well as project-specific parameters to achieve the functional requirements associated with the project design intent. In addition to the design, equipment, and system aspects of the proposed design, the design engineer should consider sustainability as it pertains to responsible energy and environmental design, as well as constructability of the design.

HVAC systems are categorized by the method used to produce, deliver, and control heating, ventilating, and air conditioning in the conditioned area. This chapter addresses procedures for selecting an appropriate system for a given application while taking into account pertinent issues associated with designing, building, commissioning, operating, and maintaining the system. It also addresses specific owner requirements and constraints associated with selecting the optimum HVAC system for the application. Chapters 2 to 18 describe specific approaches and systems along with their attributes, based on their heating and cooling medium, the commonly used variations, constructability, commissioning, operation, and maintenance.

This chapter is intended as a guide for the design engineer, builder, facility manager, and student needing to know or reference the analysis and selection process that ultimately leads to recommending the optimum system for the job. The approach applies to HVAC equipment conversions, building system upgrades, system retrofits, building renovations and expansion, and new construction for any building: small, medium, large, below grade, at grade, low-rise, and high-rise. This system analysis and selection process (Figure 1) approach helps guide the design engineer in drafting a report that recommends the best system(s) for any building program, regardless of facility type. This chapter's analysis examines objective, subjective, short-term, and long-term goals along with constraints. Figure 1 also highlights five project delivery methods: performance contracting, design-bid-build, design-build, integrated building design, and construction management.

1. SELECTING A SYSTEM

The design engineer is responsible for considering various systems and equipment and recommending one or more system options that will meet the project goals and perform per the design intent. It is imperative that the design engineer and owner collaborate to identify and prioritize criteria associated with the design goals. In addition, if the project includes preconstruction services, the designer, owner, and operator should consult with a builder to take advantage of a constructability analysis as well as the consideration of value-engineered options. Occupant comfort (as defined by ASHRAE *Standard* 55), process heating, space heating, cooling, and ventilation criteria must be considered when selecting the optimum system(s), as well as the following:

- Temperature
- Humidity
- Air motion
- Air/water velocity
- Water quality and/or reuse
- Outdoor air quality or purity
- Indoor air purity or quality
- Air changes per hour
- Acoustics and vibration
- Local climate
- Mold and mildew prevention
- Capacities (existing, proposed, and future expansion)
- Redundancy
- Spatial requirements (present and future)
- Environmental health and safety design
- Security
- First cost
- Return-on-investment cost
- Energy consumption costs
- Operator labor costs
- Maintenance costs
- Serviceability
- Reliability
- Flexibility
- Controllability
- Replaceability
- Life-cycle analysis
- Sustainability of design
- Seismic protection
- Filtration and filtration effects
- Changing codes and standards

Because these factors are interrelated, the owner, design engineer, operator, and builder must consider how these criteria affect each other. The relative importance of factors such as these varies with different owners, and can often change from one project to another for the same owner. For example, typical owner concerns include first cost compared to operating cost, extent and frequency of maintenance and whether that maintenance requires entering the occupied space, expected frequency of system failure, effect of failure, and time required to correct the failure. Each concern has a different priority, depending on the owner's goals.

Additional Goals

In addition to the primary goal of providing the desired environment, the design engineer should be aware of and account for other goals the owner may require. These goals may include the following:

The preparation of this chapter is assigned to TC 9.1, Large Building Air-Conditioning Systems.

- Seasonal start-up date
- Occupant move-in date
- Operator training
- Supporting a process, such as operation of computer equipment
- Promoting a germ-free environment
- Increasing marketability of rental spaces
- Increasing net rental income
- Increasing property saleability
- Public image of the property
- Certification of energy use (LEED®, ENERGYSTAR®, etc.)

The owner can only make appropriate value judgments if the design engineer provides complete information on the advantages and disadvantages of each system option. Just as the owner does not usually know the relative advantages and disadvantages of different HVAC systems, the design engineer rarely knows all the owner's financial and functional goals. Hence, the owner must be proactive and involved in system selection in the conceptual phase of the job. The same can be said for operator participation so that the final design intent is sustainable and can continue to fulfill design intent.

All owners should request and/or require the design team to provide building and HVAC security (as defined in Chapter 61 of the 2023 *ASHRAE Handbook—HVAC Applications*). This security, in addition to environmental health and safety criteria, should address influences outside the building (e.g., smoke, toxic fumes, flood, etc.) as well as those inside the building. System selection pertaining to HVAC security may require another design team member, a security consultant, and/or the owner's own security group.

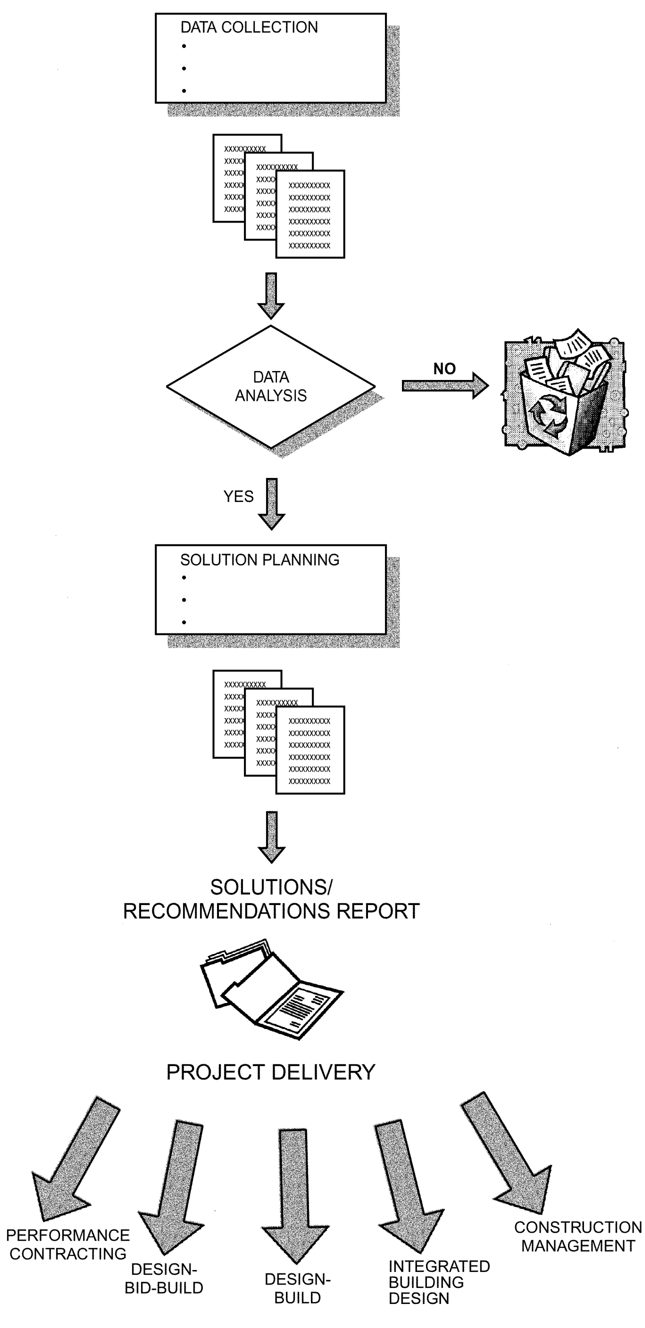

Fig. 1 Process Flow Diagram

Equipment and System Constraints

Once the goal criteria and additional goal options are verified, equipment and system constraints must be assessed and documented as part of the system analysis and selection recommendation report. These constraints may include the following:

- Performance limitations (e.g., temperature, humidity, space pressure, etc.)
- Code updates and/or new codes
- Available capacity including equipment size, as well as ductwork and/or pipe size
- Available space and access in and/or out with new or old equipment
- Available utility source
- Energy budget (e.g., conceptual Btu/h·ft^2 per year), code-driven or targeted
- Equipment efficiency
- Operator knowledge and capabilities
- Existing building occupants
- Building architecture

Few projects allow detailed quantitative evaluation of all alternatives. Common sense, historical data, and subjective experience can be used to narrow choices to one or two potential systems. Heating and air-conditioning loads often contribute to constraints, narrowing the choice to systems that fit in available space and are compatible with building architecture. Chapters 17 and 18 of the 2021 *ASHRAE Handbook—Fundamentals* describe methods to determine the size and characteristics of heating and air-conditioning loads. By establishing capacity requirements, equipment size can be determined, and the choice may be narrowed to those systems that work well on projects within the required size range.

Loads vary over time based on occupied and unoccupied periods, changes in weather, type of occupancy, activities, internal loads, and solar exposure. Each space with a different use and/or exposure may require its own control zone to maintain space comfort. Some areas with special requirements (e.g., dehumidification requirements) may need individual systems. The extent of zoning, degree of control required in each zone, and space required for individual zones also narrow system choices.

No matter how efficiently a particular system operates or how economical it is to install, it can only be considered if it (1) maintains the desired building space environment within an acceptable tolerance under expected conditions and occupant activities and (2) physically fits into, on, or adjacent to the building without causing objectionable occupancy conditions.

Cooling, heating, and humidity control are often the basis of sizing HVAC components and subsystems, but ventilation requirements can also significantly impact system sizing. For example, if large quantities of outdoor air are required for ventilation or to replace air exhausted from the building, the design engineer may only need to consider systems that transport and effectively condition those large outdoor air volumes, thus eliminating many of the other HVAC systems (e.g., unit ventilators, through-the-wall window units) from system consideration. The cooling, heating, humidification, and dehumidification delivery performance compromises may be minor for one application in a given climate, but may be unacceptable in another that has more stringent requirements.

HVAC systems and associated distribution systems often occupy a significant amount of space. Major components may also require special support from the structure. The size and appearance of terminal devices (e.g., grilles, registers, diffusers, fan-coil units, radiant panels, chilled beams) affect architectural design because they are visible in the occupied space, which constrains the design.

Sustainable energy consumption can be compromised and long-term project success lost if building operators are not trained to efficiently and effectively operate and maintain the building systems. For projects in which the design engineer used some form of energy software simulation, the resultant data should be passed on to the building owner so that goals and expectations can be measured and benchmarked against actual system performance. Even though the HVAC designer's work may be complete after system commissioning and turnover to the owner, continuous acceptable performance is expected and will only occur if the owner understands the background and systems selected and their anticipated performance. See ASHRAE *Guideline* 0 and to ASHRAE's Building Energy Quotient (bEQ) program (www.buildingeq.com).

System operability should be considered in system selection. Constructing a highly sophisticated, complex HVAC system in a building where maintenance personnel lack the required skills for maintenance can be a recipe for disaster at worst, and at best may require the use of costly outside maintenance and service contractors to achieve successful system operation.

The design engineer should closely coordinate the system constraints with the rest of the design team, as well as the owner, to overcome design obstacles associated with the HVAC systems under consideration for the project.

Constructability Constraints

The design engineer must take into account HVAC system constructability issues before the project reaches the construction document phase. Some of these constraints may significantly affect the success of the design and cannot be overlooked in the conceptual and design development phases. Some issues and concerns associated with constructability are

- Existing conditions (e.g., floor load, access into and through a building)
- Rigging equipment into and out of a building to a designated area
- Demolition and impact on adjacent space and existing systems in operation
- Maintaining existing building occupancy, use of the building, and system operation
- Ability to phase HVAC system installation
- Temporary HVAC
- Equipment availability (e.g., delivery lead time)
- Construction schedule
- Construction budget

It can be very advantageous to involve a builder into the early stages of an HVAC design to bring their prospective and experience to the project system analysis and selection process. Whether a construction manager, design-builder, or integrated project delivery leader, each brings years of experience to the topic of constructability that can influence the final selection of HVAC system(s).

Construction budget constraints can also influence the choice of HVAC systems. Based on historical data, some systems may not be economically viable within the budget limitations of an owner's building program. In addition, annual maintenance and operating budget (utilities, labor, and materials) can be limiting although more often than not, annual operating budgets do not get included in the total construction "soft costs" (e.g., furnishing, general conditions). This can be particularly important for building owners who will retain the building for a substantial number of years.The building's estimator can assist in the overall project cost by introducing value-engineered solutions that may involve (1) cost-driven performance, which may provide a better solution for lower first cost; (2) a more sustainable solution over the life of the equipment; or (3) best value based on a reasonable return on investment.

Narrowing the Choices

The following chapters in this volume present information to help the design engineer narrow the choices of HVAC systems:

- Chapter 2 focuses on a distributed approach to HVAC.
- Chapter 3 provides guidance for large equipment centrally located in or adjacent to a building.
- Chapters 4 to 18 address the numerous types of air-conditioning and heating systems.

Each chapter summarizes positive and negative features of various systems. Comparing the criteria, other factors and constraints, and their relative importance usually identifies one or two systems that best satisfy project goals. In making choices, notes should be kept on all systems considered and the reasons for eliminating those that are unacceptable.

Each selection may require combining a primary system with a secondary (or distribution) system. The primary system converts energy derived from fuel or electricity to produce a heating and/or cooling medium. The secondary system delivers heating, ventilation, cooling, humidification, and/or dehumidification to the occupied space. The systems are independent to a great extent, so several secondary systems may work with a particular primary system. In some cases, however, only one secondary system may be suitable for a particular primary system.

Once subjective analysis has identified one or more HVAC systems (sometimes only one choice remains), detailed quantitative evaluations must be made. All systems considered should provide satisfactory performance to meet the owner's essential goals. The design engineer should provide the owner with specific data on each system to make an informed choice. Consult the following chapters to help narrow the choices:

- Chapter 10 of the 2021 *ASHRAE Handbook—Fundamentals* covers physiological principles, comfort, and health.
- Chapter 19 of the 2021 *ASHRAE Handbook—Fundamentals* covers methods for estimating annual energy costs.
- Chapters 37 to 44 of the 2023 *ASHRAE Handbook—HVAC Applications* address important topics pertinent to the long-term success of building programs.

Other documents and guidelines that should be consulted are ASHRAE standards and guidelines; local, state, and federal guidelines; and special agency requirements (e.g., U.S. General Services Administration [GSA], Food and Drug Administration [FDA], Joint Commission on Accreditation of Healthcare Organizations [JCAHO], Facility Guidelines Institute [FGI], Leadership in Energy and Environmental Design [LEED®]). Additional sources of detailed information are listed in the Bibliography and/or available in the ASHRAE Bookstore (www.ashrae.org/bookstore).

Selection Report

As the last step, the design engineer should prepare a system analysis and selection report in a format that incorporates the following:

- Introduction
- Building program primary HVAC goals
- Criteria for selection
- Important factors, including advantages and disadvantages

Table 1 Sample HVAC System Analysis and Selection Matrix (0 to 10 Score)

Goal: Furnish and install an HVAC system that provides moderate space temperature control with minimum humidity control at an operating budget of 70,000 Btu/h per square foot per year				
Categories	**System #1**	**System #2**	**System #3**	**Remarks**
1. Criteria for selection: • 78°F space temperature with ±3°F control during occupied cycle, with 40% rh and ±5% rh control during cooling. • 68°F space temperature with ±2°F, with 20% rh and ±5% rh control during heating season. • First cost • Equipment life cycle				
2. Important factors: • First-class office space stature • Individual tenant utility metering				
3. Other goals: • Engineered smoke control system • ASHRAE *Standard* 62.1 ventilation rates • Direct digital control building automation				
4. System constraints: • No equipment on first floor • No equipment on ground adjacent to building				
5. Energy use as predicted by use of an industry-acceptable computerized energy model				
6. Other constraints: • No perimeter finned-tube radiation or other type of in-room equipment				
TOTAL SCORE				

- System integration with other building systems
- Other HVAC goals
- Security and environmental health and safety criteria
- Building envelope
- Project timeline (design, equipment delivery, commissioning, training, and construction)
- Basis of design
- HVAC system analysis and selection matrix
- System narratives
- Budget costs (first cost, operating cost, and energy cost)
- Final recommendation(s)

A brief outline of each of the final selections should be provided along with overall budget and/or cost per square foot. In addition, HVAC systems deemed inappropriate should be noted as having been considered but not found applicable to meet the owner's primary HVAC goal.

The report should include an HVAC system selection matrix that identifies the one or two suggested HVAC system selections (primary and secondary, when applicable), system constraints, and other constraints and considerations. In completing this matrix assessment, the design engineer should identify the owner's input to the analysis. This input can also be applied as weighted multipliers, because not all criteria carry the same weighted value.

Many grading methods are available to complete an analytical matrix analysis. Probably the simplest is to rate each item excellent, very good, good, fair, or poor. A numerical rating system such as 0 to 10, with 10 equal to excellent and 0 equal to poor or not applicable, can provide a quantitative result. The HVAC system with the highest numerical value then becomes the recommended system to accomplish the goal.

The system selection report should include a summary followed by a more detailed account of the HVAC system analysis and system selection. This summary should highlight key points and findings that led to the recommendation(s). The analysis should refer to the system selection matrix (such as in Table 1) and the reasons for scoring.

With each HVAC system considered, the design engineer should note the criteria associated with each selection. Issues such as close-tolerance temperature and humidity control may eliminate some HVAC systems from consideration. System constraints, noted with each analysis, may eliminate potential HVAC systems. Advantages and disadvantages of each system should be noted with the scoring from the HVAC system selection matrix. This process should reduce HVAC selection to one or two optimum choices for presentation to the owner. Examples of similar installations for other owners can be included with this report to support the final recommendation. Identifying a third party for an endorsement allows the owner to inquire about the success of other HVAC installations.

2. HVAC SYSTEMS AND EQUIPMENT

Many built, expanded, and/or renovated buildings may be ideally suited for decentralized HVAC systems, with equipment located in, throughout, adjacent to, or on top of the building. The alternative to this decentralized approach is to use primary equipment located in a central plant (either inside or outside the building) with water and/or air required for HVAC needs distributed from this plant.

Decentralized System Characteristics

The various types of decentralized systems are described in Chapter 2. The common element is that the required cooling is distributed throughout the building, generally with direct-expansion cooling, electric, or gas-fired heating of air systems, and on occasion, oil-fired warm-air heating units.

Temperature, Humidity, and Space Pressure Requirements. A decentralized system may be able to fulfill any or all of these design parameters, but typically not as efficiently or as accurately as a central system.

Capacity Requirements. A decentralized system usually requires each piece of equipment to be sized for zone peak capacity, unless the systems are variable-volume. Depending on equipment type and location, decentralized systems do not benefit as much from equipment sizing diversity as centralized systems do.

Redundancy. A decentralized system may not have the benefit of back-up or standby equipment. This limitation may need review.

Facility Management. A decentralized system can allow the building manager to maximize performance using good business/facility management techniques in operating and maintaining the HVAC equipment and systems. With some decentralized equipment (e.g., rooftop HVAC units), operation and maintenance may be performed annually via a service contract with a mechanical contracting company.

Spatial Requirements. A decentralized system may or may not require equipment rooms. Because of space restrictions imposed on the design engineer or architect, equipment may be located on the roof and/or the ground adjacent to the building. Depending on system components, additional space may be required in the building for chillers and boilers. Likewise, a decentralized system may or may not require duct and pipe shafts throughout the building.

First Cost. A decentralized system probably has the best first-cost benefit. This feature can be enhanced by phasing in the purchase of decentralized equipment as needed (i.e., buying equipment as the building is being leased/occupied).

Operating Cost. A decentralized system can save operating cost by strategically starting and stopping multiple pieces of equipment. When comparing energy consumption based on peak energy draw, decentralized equipment may not be as attractive as larger, more energy-efficient centralized equipment.

Maintenance Cost. A decentralized system can save maintenance cost when equipment is conveniently located and equipment size and associated components (e.g., filters) are standardized. When equipment is located outdoors, maintenance may be difficult during bad weather.

Reliability. A decentralized system usually has reliable equipment, although the estimated equipment service life may be less than that of centralized equipment. Decentralized system equipment may, however, require maintenance in the occupied space.

Flexibility. A decentralized system may be very flexible because it may be placed in numerous locations.

Level of Control. Decentralized systems often use direct refrigerant expansion (DX) for cooling, and on/off or staged heat. This step control results in greater variation in space temperature and humidity, where close control is not desired or necessary. As a caution, oversizing DX or stepped cooling can allow high indoor humidity levels and mold or mildew problems.

Noise and Vibration. Decentralized systems often locate noisy machinery close to building occupants.

Constructability. Decentralized systems frequently consist of multiple and similar-in-size equipment that makes standardization a construction feature, as well as purchasing units in large quantities.

Centralized System Characteristics

These systems are characterized by central refrigeration equipment/systems with chilled-water distribution and central boiler equipment/system with hot-water distribution. Other central systems include low-, medium-, and high-pressure steam plants and cogeneration central equipment/systems, or condenser water systems used with water-source heat pumps.

Among the largest centralized systems are HVAC plants serving groups of large buildings. These plants improve diversity and generally operate more efficiently, and with lower maintenance costs, than individual central plants. Economic considerations of larger centralized systems require extensive analysis. The utility analysis may consider multiple fuels and may also include gas and steam turbine-driven equipment. Multiple types of primary equipment using multiple fuels and types of HVAC-generating equipment (e.g., centrifugal and absorption chillers) may be combined in one plant.

Temperature, Humidity, and Space Pressure Requirements. A central system may be able to fulfill any or all of these design parameters, typically with greater precision and efficiency than a decentralized system.

Capacity Requirements. A central system usually allows the design engineer to consider HVAC diversity factors that significantly reduce installed equipment capacity. As a result, this offers some attractive first-cost and operating-cost benefits.

Redundancy. A central system can accommodate standby equipment that decentralized configurations may have trouble accommodating because of cost.

Facility Management. A central system usually allows the building manager to maximize performance using good business/facility management techniques in operating and maintaining the HVAC equipment and systems. Operation and maintenance are usually handled by on-site staff, whether employees of the building owner or outsourced building management placed on site.

Spatial Requirements. The equipment room for a central system is normally located outside the conditioned area: in a basement, penthouse, service area, or adjacent to or remote from the building. A disadvantage of this approach may be the additional cost to furnish and install secondary equipment for the air and/or water distribution. Other considerations include access requirements and physical constraints that exist throughout the building to the installation of the secondary distribution network of ducts and/or pipes and for equipment replacement.

First Cost. Even with HVAC diversity, a central system may not be less costly than decentralized HVAC systems. Historically, central system equipment has a longer equipment service life to compensate for this shortcoming. Thus, a life-cycle cost analysis is very important when evaluating central versus decentralized systems.

Operating Cost. A central system usually has the advantage of larger, more energy-efficient primary equipment compared to decentralized system equipment. In addition, the availability of multiple pieces of HVAC equipment allows staging of this equipment operation to match building loads while maximizing operational efficiency.

Maintenance Cost. The equipment room for a central system provides the benefit of being able to maintain HVAC equipment away from occupants in an appropriate service work environment. Access to occupant workspace is not required, thus eliminating disruption to the space environment, product, or process. Because of the typically larger capacity of central equipment, there are usually fewer pieces of HVAC equipment to service.

Reliability. Centralized system equipment generally has a longer service life.

Flexibility. Flexibility can be a benefit when selecting equipment that provides an alternative or back-up source of HVAC.

Air Distribution Systems

The various air distribution systems, including dedicated outdoor air systems (DOAS), are detailed in Chapter 4; DOAS is also covered more thoroughly in Chapter 51. Any of the preceding system types discussed within this chapter can be used in conjunction with DOAS.

Level of Control. Centralized air systems generally use chilled water for cooling, and steam or hot water for heating. This usually allows for close control of space temperature and humidity where desired or necessary.

Sound and Vibration. Centralized air system placement should consider the associated equipment noise and vibration and locate machinery sufficiently remote from building occupants or noise-sensitive processes.

Constructability. Centralized air systems usually require more coordinated installation than decentralized systems. When reusing an existing central air fan(s) and associated ductwork, it is important to verify construction class remains the same or is reduced (e.g., increasing supply air fan capacity in a project renovation may increase the fan construction from a Class 1 to a Class 2 construction). The same can be said when increasing airflow within an existing duct distribution system (e.g., ductwork may require reinforcement and/or duct seams sealed airtight due to the increase in duct static and velocity pressures).

Primary Equipment

The type of decentralized and centralized equipment selected for buildings depends on a well-organized HVAC analysis and selection report. The choice of primary equipment and components depends on factors presented in the selection report (see the section on Selecting a System). Primary HVAC equipment includes refrigeration equipment; heating equipment; and air, water, and steam delivery equipment.

Many HVAC designs recover internal heat from lights, people, and equipment to reduce the size of the heating plant. In buildings with core areas that require cooling while perimeter areas require heating, one of several heat reclaim systems can heat the perimeter to save energy. Sustainable design is also important when considering recovery and reuse of materials and energy. Chapter 9 describes heat pumps and some heat recovery arrangements, Chapter 37 describes solar energy equipment, and Chapter 26 introduces air-to-air energy recovery. In the 2023 *ASHRAE Handbook—HVAC Applications*, Chapter 37 covers energy management and Chapter 42 covers building energy monitoring. Chapter 35 of the 2021 *ASHRAE Handbook—Fundamentals* provides information on sustainable design, and Chapters 10 and 16 of that volume address indoor air quality, ventilation, and infiltration in the context of assessing optimum HVAC application options.

The search for energy savings has extended to **cogeneration** or **total energy (combined heat and power [CHP])** systems, in which on-site power generation is added to the HVAC project. The economic viability of this function is determined by the difference between gas and electric rates and by the ratio of electricity to heating demands for the project. In these systems, waste heat from generators can be transferred to the HVAC systems (e.g., to drive turbines of centrifugal compressors, serve an absorption chiller, provide heating or process steam). Chapter 7 covers cogeneration or total energy systems. Alternative fuel sources, such as waste heat boilers, are now included in fuel evaluation and selection for HVAC applications.

Thermal storage is another cost-saving concept, which provides the possibility of off-peak generation of chilled water or ice. Thermal storage can also be used for storing hot water for heating. Many electric utilities impose severe charges for peak summer power use or offer incentives for off-peak use. Storage capacity installed to level the summer load may also be available for use in winter, thus making heat reclaim a viable option. Chapter 50 has more information on thermal storage.

With ice storage, colder supply air can be provided than that available from a conventional chilled-water system. This colder air allows use of smaller fans and ducts, which reduces first cost and (in some locations) operating cost. Additional pipe and duct insulation is often required, however, contributing to a higher first cost. These life-cycle savings can offset the first cost for storage provisions and the energy cost required to make ice. Similarly, thermal storage of hot water can be used for heating.

Refrigeration Equipment

Chapters 2 and 3 summarize the primary types of refrigeration equipment for HVAC systems.

When chilled water is supplied from a central plant, as on university campuses and in downtown areas of large cities or in large buildings, the utility service provider should be contacted during system analysis and selection to determine availability, cost, and the specific requirements of the service.

Heating Equipment

Steam boilers and heating-water boilers are the primary means of heating a space using a centralized system, as well as some decentralized systems. These boilers may be (1) used both for comfort and process heating; (2) manufactured to produce high- or low-pressure steam; and (3) fired with coal, oil, electricity, gas, and even some waste materials. Low-pressure boilers are rated for a working pressure of 15 psig for steam, and 160 psig for water, with a temperature limit of 250°F. Packaged boilers, with all components and controls assembled at the factory as a unit, are available. Electrode or resistance electric boilers that generate either steam or hot water are also available. Chapter 32 has further information on boilers, and Chapter 27 details air-heating coils.

Where steam, hot water, or chilled water is supplied from a central plant, as on university campuses and in downtown areas of large cities, the utility provider should be contacted during project system analysis and selection to determine availability, cost, and specific requirements of the service.

When primary heating equipment is selected, the fuels considered must ensure maximum efficiency. Chapter 31 discusses design, selection, and operation of the burners for different types of primary heating equipment. Chapter 28 of the 2021 *ASHRAE Handbook—Fundamentals* describes types of fuel, fuel properties, and proper combustion factors.

Air Delivery Equipment

Primary air delivery equipment for HVAC systems is classified as packaged, manufactured and custom-manufactured, or field-fabricated (built-up). Most air delivery equipment for large systems uses centrifugal or axial fans; however, plug or plenum fans are often used. Centrifugal fans are frequently used in packaged and manufactured HVAC equipment. One system rising in popularity is a fan array, which uses multiple plug fans on a common plenum wall, thus reducing unit size. Axial fans are more often part of a custom unit or a field-fabricated unit. Both types of fans can be used as industrial process and high-pressure blowers. Chapter 21 describes fans, and Chapters 19 and 20 provide information about air delivery components.

3. SPACE REQUIREMENTS

In the initial phase of building design, the design engineer seldom has sufficient information to render the optimum HVAC design for the project, and its space requirements are often based on percentage of total area or other experiential rules of thumb. The final design is usually a compromise between the engineer's recommendations and the architectural considerations that can be accommodated in the building. An integrated project design (IPD) approach, as recommended by the American Institute of Architects (AIA), can address these problems early in the design process; also see Chapter 60 of the 2023 *ASHRAE Handbook—HVAC Applications*. Other times, the building owner may prefer either a centralized or decentralized system and dictate final design and space requirements. This section discusses some of these requirements.

Equipment Rooms

Total mechanical and electrical space requirements range between 4 and 9% of gross building area, with most buildings in the 6 to 9% range. These ranges include space for HVAC, electrical, plumbing, and fire protection equipment and may also include vertical shaft space for mechanical and electrical distribution through the building.

Most equipment rooms should be centrally located to (1) minimize long duct, pipe, and conduit runs and sizes; (2) simplify shaft layouts; and (3) centralize maintenance and operation. With shorter duct and pipe runs, a central location could also reduce pump and fan motor power requirements, which reduces building operating costs. But, for many reasons, not all mechanical and electrical equipment rooms can be centrally located in the building. In any case, equipment should be kept together whenever possible to minimize space requirements, centralize maintenance and operation, and simplify the electrical system.

Equipment rooms generally require clear ceiling height ranging from 10 to 18 ft, depending on equipment sizes and the complexity of air and/or water distribution.

The main electrical transformer and switchgear rooms should be located as close to the incoming electrical service as practical. If there is an emergency generator, it should be located considering (1) proximity to emergency electrical loads and sources of combustion and cooling air and fuel, (2) ease of properly venting exhaust gases to the outdoors, and (3) provisions for noise control. Noise control is generally a minor issue because equipment only runs during emergencies.

Primary Equipment Rooms. The heating equipment room houses the boiler(s) and possibly a boiler feed unit (for steam boilers), chemical treatment equipment, pumps, heat exchangers, pressure-reducing equipment, air compressors, and miscellaneous ancillary equipment. The refrigeration equipment room houses the chiller(s) and possibly chilled-water and condenser water pumps, heat exchangers, air-conditioning equipment, air compressors, and miscellaneous ancillary equipment, and in some cases, direct-expansion compressors. Design of these rooms needs to consider (1) equipment size and weight, (2) installation, maintenance, and replacement, (3) applicable regulations relative to combustion air and ventilation air, and (4) noise and vibration transmission to adjacent spaces. Consult ASHRAE *Standard* 15 for refrigeration equipment room safety requirements. Note that primary equipment rooms should not be located below ground or even at ground level if the project is to be located in a designated flood zone area, due to the potential of the equipment room being flooded.

Many air-conditioned buildings require a cooling tower or other type of heat rejection equipment. If the cooling tower or water-cooled condenser is located at ground level, it should be at least 100 ft away from the building to reduce tower noise in the building, and to keep discharge air and moisture carryover from fogging the building's windows and discoloring the building facade or from contaminating outdoor air being introduced into the building. Cooling towers should be kept a similar distance from parking lots to avoid staining car finishes with atomized water treatment chemicals. Chapter 40 has further information on this equipment.

It is often economical to locate the heating and/or refrigeration plant in the building, on an intermediate floor, in a roof penthouse, or on the roof when roof space is available; air-cooled refrigeration systems can be considered. Electrical service and structural costs are higher, but these may be offset by reduced costs for piping, pumps and pumping energy, and chimney requirements for fuel-fired boilers. Also, initial cost of equipment in a tall building may be less for equipment located on a higher floor (e.g., boilers located in a roof penthouse of a high-rise building) because some operating pressures may be lower.

Regulations applicable to both gas and fuel oil systems must be followed. Gas fuel may be more desirable than fuel oil because of the physical constraints on the required fuel oil storage tank, as well as specific environmental and safety concerns related to oil leaks. Also, the cost of an oil leak detection and prevention system may be substantial. Oil pumping presents added design and operating problems, depending on location of the oil tank relative to the oil burner.

Energy recovery systems can reduce the size of the heating and/or refrigeration plant. The possibility of failure or the need to take heat recovery equipment out of operation for service should be considered in the heating plant design, to ensure the ability to heat the building with the heating plant without heat recovery. Well-insulated buildings and electric and gas utility rate structures may encourage the design engineer to consider energy conservation concepts such as limiting demand, ambient cooling, and thermal storage.

Fan Rooms

Fan rooms house HVAC air delivery equipment and may include other miscellaneous equipment. The room must have space for removing the fan(s), shaft(s), coils, and filters. Installation, replacement, and maintenance of this equipment should be considered when locating and arranging the room.

Fan rooms in a basement that has an airway for intake of outdoor air present a potential problem. Low air intakes are a security concern, because harmful substances could easily be introduced (see the section on Security). Debris, and in some climate zones snow, may fill the area, resulting in safety, health, and fan performance concerns. Parking areas close to the building's outdoor air intake may compromise ventilation air quality.

Fan rooms located at the perimeter wall can have intake and exhaust louvers at the location of the fan room, subject to coordination with architectural considerations. Interior fan rooms often require intake and exhaust shafts from the roof because of the difficulty (typically caused by limited ceiling space) in ducting intake and exhaust to the perimeter wall.

Fan rooms on the second floor and above have easier access for outdoor and exhaust air. Depending on the fan room location, equipment replacement may be easier. The number of fan rooms required depends largely on the total floor area and whether the HVAC system is centralized or decentralized. Buildings with large floor areas may have multiple decentralized fan rooms on each or alternate floors. High-rise buildings may opt for decentralized fan rooms for each floor, or for more centralized service with one fan room serving the lower 10 to 20 floors, one serving the middle floors of the building, and one at the roof serving the top floors.

Life safety is a very important factor in HVAC fan room location. Chapter 54 of the 2023 *ASHRAE Handbook—HVAC Applications* discusses fire and smoke control. State and local codes have additional fire and smoke detection and damper criteria. Building codes that require smoke control and building pressurization should be accounted for in equipment selection.

Horizontal Distribution

Most decentralized and central systems rely on horizontal distribution. To accommodate this need, the design engineer needs to take into account the duct and/or pipe distribution criteria for installation in a ceiling space or below a raised floor space. Systems using water distribution usually require the least amount of ceiling or raised floor depth, whereas air distribution systems have the largest demand for horizontal distribution height. Steam systems need to accommodate pitch of steam pipe, end of main drip, and condensate return pipe pitch. Another consideration in the horizontal space cavity is accommodating the structural members, light fixtures, rain leaders, cable trays, etc., that can fill up this space.

Vertical Shafts

Buildings over two stories high usually require vertical shafts to consolidate mechanical (e.g., air distribution ducts and pipes), electrical, plumbing, and telecommunication distribution through the facility.

Air distribution includes HVAC supply, return, and exhaust air ductwork. If a shaft is used as a return air plenum, close coordination with the architect is necessary to ensure that the shaft is airtight. If the shaft is used to convey outdoor air to decentralized systems, close coordination with the architect is also necessary to ensure that the shaft is constructed to meet mechanical code requirements and to accommodate the anticipated internal pressure. The temperature of air in the shaft must be considered when using the shaft enclosure to convey outdoor air. Pipe distribution includes heating, chilled, and condenser water, and steam supply and condensate return. It may also include fuel oil and gas pumping. Other distribution systems found in vertical shafts or located vertically in the building include electric conduits/bus ducts/closets, telephone cabling/closets, uninterruptible power supply (UPS), plumbing, fire protection piping, pneumatic tubes, and conveyors.

Vertical shafts should not be adjacent to stairs, electrical closets, and elevators unless at least two sides are available to allow access to ducts, pipes, and conduits that enter and exit the shaft while allowing maximum headroom at the ceiling. In general, duct shafts with an aspect ratio of 2:1 to 4:1 are easier to develop than large square shafts. The rectangular shape also facilitates transition from the equipment in the fan rooms to the shafts. In addition, significant space is required for motorized and fire/smoke dampers when ducts leave shafts.

In multistory buildings, a central vertical distribution system with minimal horizontal branch ducts is desirable. This arrangement (1) is usually less costly; (2) is easier to balance; (3) creates less conflict with pipes, beams, and lights; and (4) enables the architect to design lower floor-to-floor heights. These advantages also apply to vertical water and steam pipe distribution systems.

The number of shafts is a function of building size and shape. In larger buildings, it is usually more economical in cost and space to have several small shafts rather than one large shaft. Separate HVAC supply, return, and exhaust air duct shafts may be desirable to reduce the number of duct crossovers. The same applies for steam supply and condensate return pipe shafts because the pipe must be pitched in the direction of flow.

When future expansion is a consideration, a pre-agreed percentage of additional shaft space should be considered for inclusion. This, however, affects the building's first cost because of the additional floor space that must be constructed. The need for access doors into shafts and gratings at various locations throughout the height of the shaft should also be considered.

Rooftop Equipment

For buildings three stories or less, system analysis and selection frequently locates HVAC equipment on the roof or another outdoor location, where the equipment is exposed to the weather. Decentralized equipment and systems are sometimes more advantageous than centralized HVAC for smaller buildings, particularly those with multiple tenants with different HVAC needs. Selection of rooftop equipment is usually driven by first cost versus operating cost and/or maximum service life of the equipment. For buildings with larger floor plates, centralized equipment may be advantageous and should be considered.

Equipment Access

Properly designed mechanical and electrical equipment rooms must allow for movement of large, heavy equipment in, out, and through the building. If a building is being renovated for another application, floor loading should be assessed to determine whether heavier equipment can be installed on the existing floor. Equipment replacement and maintenance can be very costly if access is not planned properly. Access to rooftop equipment should be by means of, at a minimum, a ship's ladder and not by a vertical ladder, taking into account a technician carrying boxes of replacement filters. Use caution when accessing equipment on sloped roofs.

Because systems vary greatly, it is difficult to estimate space requirements for refrigeration and boiler rooms without making block layouts of the system selected. Block layouts allow the design engineer to develop the most efficient arrangement of the equipment with adequate access and serviceability. Block layouts can also be used in preliminary discussions with the owner and architect. Only then can the design engineer verify the estimates and provide a workable and economical design.

4. AIR DISTRIBUTION

Ductwork should deliver conditioned air to an area as directly, quietly, and economically as possible. Structural features of the building generally require some compromise and often limit the depth of space available for ducts. Chapter 10 discusses air distribution design for small heating and cooling systems. Chapters 20 and 21 of the 2021 *ASHRAE Handbook—Fundamentals* discuss space air distribution and duct design.

The design engineer must coordinate duct layout with the structure as well as other mechanical, electrical, plumbing, and communication systems. In commercial projects, the design engineer is usually encouraged to reduce floor-to-floor dimensions: with the resulting decrease in available interstitial space, depth for ducts is a major design challenge. In institutional and information technology buildings, higher floor-to-floor heights are required because of the sophisticated, complex mechanical, electrical, and communication distribution systems.

Exhaust systems, especially those serving fume exhaust, grease exhaust, dust and/or particle collection, and other process exhaust, require special design considerations. Capture velocity, duct material, and pertinent duct fittings and fabrication are a few of the design parameters necessary for this type of distribution system to function properly, efficiently, and per applicable codes. See Chapters 15 to 33 of the 2023 *ASHRAE Handbook—HVAC Applications* for additional information on industrial applications.

Air Terminal Units

In some instances, such as in low-velocity, all-air systems, air may enter from the supply air ductwork directly into the conditioned space through a grille, register, or diffuser. In medium- and high-velocity air systems, an intermediate device normally controls air volume, reduces air pressure from the duct to the space, or both. Various types of air terminal units are available, including (1) a fan-powered terminal unit, which uses an integral fan to mix ceiling plenum air and primary air from the central or decentralized fan system rather than depending on induction (mixed air terminal unit) delivering to low-pressure ductwork and then to the space); (2) a variable-air-volume (VAV) terminal unit, which varies the amount of air delivered to the space (this air may be delivered to low-pressure ductwork and then to the space, or the terminal may contain an integral air diffuser); or (3) other in-room terminal type (see Chapter 5). Chapter 20 has more information about air terminal units.

See Chapters 20 and 21 of the 2021 *ASHRAE Handbook—Fundamentals* for information on space air diffusion and duct design.

Duct Insulation

In new construction and renovation upgrade projects, HVAC supply air ducts should be insulated and sealed in accordance with energy code requirements. ASHRAE *Standard* 90.1 and Chapter 23 of the 2021 *ASHRAE Handbook—Fundamentals* have more information about insulation and calculation methods.

Ceiling and Floor Plenums

Frequently, the space between the suspended ceiling and the floor slab above it is used as a return air plenum to reduce distribution ductwork. Check regulations before using this approach in new construction or renovation because most codes prohibit combustible material in a ceiling return air plenum. Ceiling plenums and raised floors can also be used for supply air displacement systems to minimize horizontal distribution, along with other features discussed in Chapter 4.

Some ceiling plenum applications with lay-in panels do not work well where the stack effect of a high-rise building or high-rise elevators creates a negative pressure. If the plenum leaks to the low-pressure area, tiles may lift and drop out when the outside door is opened and closed.

The air temperature in a return air plenum directly below a roof deck is usually higher by 3 to 5°F during the air-conditioning season than in a ducted return. This can be an advantage to the occupied space below because heat gain to the space is reduced. Conversely, return air plenums directly below a roof deck have lower return air temperatures during the heating season than a ducted return and may require supplemental heat in the plenum.

Using a raised floor for air distribution is popular for computer rooms and cleanrooms, and is now being used in other HVAC applications. Underfloor air displacement (UFAD) systems in office buildings use the raised floor as a supply air plenum, which could reduce overall first cost of construction and ongoing improvement costs for occupants. Making the underfloor supply air plenum as airtight as possible is critical to the success of this type of plenum. This UFAD system improves air circulation to the occupied area of the space. See Chapter 20 of the 2023 *ASHRAE Handbook—HVAC Applications* and Chapter 20 of the 2021 *ASHRAE Handbook—Fundamentals* for more information on displacement ventilation and underfloor air distribution.

5. PIPE DISTRIBUTION

Piping should deliver refrigerant, heating water, chilled water, condenser water, fuel oil, gas, steam, and condensate drainage and return to and from HVAC equipment as directly, quietly, and economically as possible. Structural features of the building generally require mechanical and electrical coordination to accommodate P-traps, pipe pitch-draining of low points in the system, and venting of high points. When assessing application of pipe distribution to air distribution, the floor-to-floor height requirement can influence the pipe system: it requires less ceiling space to install pipe when compared to air ductwork. An alternative to horizontal piping is vertical pipe distribution, which may further reduce floor-to-floor height criteria. See Chapter 22 of the 2021 *ASHRAE Handbook—Fundamentals* for pipe sizing.

Pipe Systems

HVAC piping systems can be divided into two parts: (1) piping in the central plant equipment room and (2) piping required to deliver refrigerant, heating water, chilled water, condenser water, fuel oil, gas, steam, and condensate drainage and return to and from decentralized HVAC and process equipment throughout the building. Chapters 11 to 15 discuss piping for various heating and cooling systems. Chapters 1 to 4 of the 2022 *ASHRAE Handbook—Refrigeration* discuss refrigerant system design and practices.

Pipe Insulation

In new construction and renovation projects, most HVAC piping must be insulated. ASHRAE *Standard* 90.1 and Chapter 23 of the 2021 *ASHRAE Handbook—Fundamentals* have information on insulation and calculation methods. In most applications, water-source heat pump loop piping and condenser water piping may not need to be insulated.

6. SECURITY AND ENVIRONMENTAL HEALTH AND SAFETY

Since September 11, 2001, much attention has been given to protecting buildings against terrorist attacks via their HVAC systems. The first consideration should be to perform a risk assessment of the particular facility, which may be based on usage, size, population, and/or significance. The risk assessment is a subjective judgment by the building owner (and sometimes by a registered/certified security professional) of whether the building is at low, medium, or high risk. An example of low-to-medium risk buildings may be suburban office buildings, shopping malls, hospitals, educational institutions, or major office buildings. High-risk buildings may include major government buildings. The level of protection designed into these buildings may include enhanced particulate filtration, gaseous-phase filtration, and various control schemes to allow purging of the facility using either the HVAC system or an independent dedicated system.

Enhanced particulate filtration for air-handling systems to the level of MERV 14 to 16 filters not only tends to reduce circulation of dangerous substances (e.g., anthrax), but also provides better indoor air quality (IAQ). Gaseous-phase filtration can remove harmful substances such as sarin and other gaseous threats. For buildings of all risk levels, consideration should include proper location of outdoor air intakes to account for other security concerns, such as toxic fumes from a local accident (e.g., punctured tank from a nearby railroad car or tanker truck). The extent to which the HVAC system designer should use these measures could depend on the recommendations of an experienced security and environmental health and safety consultant or agency.

In any building, consideration should be given to protecting outdoor air intakes against insertion of dangerous materials by locating the intakes on the roof or substantially above grade level. Separate systems for mailrooms, loading docks, and other similar spaces should be considered so that any dangerous material received cannot be spread throughout the building from these types of vulnerable spaces. Emergency ventilation systems for these types of spaces should be designed so that upon detection of suspicious material, toxic fumes, or chemical spill, these spaces can be quickly purged and maintained at a negative pressure.

A more extensive discussion of this topic can be found in ASHRAE's *Guideline* 29. Note, however, that fan energy required to move air through these high efficiency filters is very significant.

7. AUTOMATIC CONTROLS AND BUILDING MANAGEMENT SYSTEMS

Basic HVAC system controls are primarily direct digital controls (DDC), but there are also electric, electronic, and pneumatic controls (found largely as part of existing building HVAC systems). Depending on the application, the design engineer may recommend a simple and basic system strategy as a cost-effective solution to an owner's heating, ventilation, and cooling needs. Chapter 48 of the 2023 *ASHRAE Handbook—HVAC Applications* and Chapter 7 of the 2021 *ASHRAE Handbook—Fundamentals* discuss automatic control in more detail.

The next level of HVAC system management is DDC wireless systems. Many DDC installations will use either pneumatic or electric control damper and valve actuators. This automatic control enhancement may include energy monitoring and energy management software. Controls may also be accessible by the building manager using a modem to a remote computer at an off-site location or via a smartphone or tablet computer with Internet connection to the control system. Building size has little to no effect on modern computerized controls: programmable controls can be furnished on the smallest HVAC equipment for the smallest projects. Chapter 43 of the 2023 *ASHRAE Handbook—HVAC Applications* covers building operating dynamics.

Automatic controls often come prepackaged and prewired on the HVAC equipment, creating a design challenge to have the equipment communicate with a third party building automation system (BAS). Current HVAC controls and their capabilities need to be compatible with other new and existing automatic controls. Chapter 41 of the 2023 *ASHRAE Handbook—HVAC Applications* discusses computer applications, and ASHRAE *Standard* 135 discusses interfacing building automation systems. Furthermore, compatibility with BACnet® and/or LonWorks® provides an additional level of compatibility between equipment made by numerous manufacturers.

Using computers and proper software, the design engineer and building manager can provide complete facility management, as well as remote facility management. A comprehensive building management system (BMS) may include HVAC system control, energy management, operation and maintenance management, medical gas system monitoring, fire alarm system, security system, lighting control, and other reporting and trending software. This system may also be integrated and accessible from the owner's computer network and the Internet.

The automatic control system can be as simple as a time clock to start and stop equipment, or as sophisticated as computerized building automation serving a decentralized HVAC system, multiple building systems, central plant system, and/or a large campus. With a focus on energy management, the building management system can be an important business tool in achieving sustainable facility management that begins with using the system selection matrix.

8. MAINTENANCE MANAGEMENT

Whereas BMS focus on operation of HVAC, electrical, plumbing, and other systems, maintenance management systems focus on maintaining assets, which include mechanical and electrical systems along with the building structure. A rule of thumb is that only about 20% of the cost of the building is in the first cost, with the other 80% being operation, maintenance, and rejuvenation of the building and building systems over the life cycle. When considering the optimum HVAC selection and recommendation at the start of a project, a maintenance management system should be considered for HVAC systems with an estimated long useful service life.

Another maintenance management business tool is a **computerized maintenance management software (CMMS)** system. The CMMS system can include an equipment database, parts and material inventory, project management software, labor records, etc., pertinent to sustainable management of the building over its life. CMMS also can integrate computer-aided drawing (CAD), building information modeling (BIM), digital photography and audio/video systems, equipment run-time monitoring and trending, and other proactive facility management systems.

In scoring the HVAC system selection matrix, consideration should also be given to the potential for interface of the BMS with the CMMS. In addition, onboard equipment diagnostics and remote troubleshooting capabilities as part of considering the optimum HVAC selection.

Planning in the design phase and the early compilation of record documents (e.g., computer-aided drawing and electronic text files, checklists, digital photos taken during construction) are also integral to successful building management and maintenance. The use of bar coding equipment, valves, etc., can link them to the CMMS via an Internet-accessed scanner.

9. BUILDING SYSTEM COMMISSIONING

When compiling data to complete the HVAC system selection matrix to analytically determine the optimum HVAC system for the project, a design engineer should begin to produce the design intent document/basis of design that identifies the project goals. This process is the beginning of building system commissioning and should be an integral part of the project documentation. As design progresses and the contract documents take shape, the commissioning process continues to be built into what eventually becomes the final commissioning report, approximately one year after the construction phase has been completed and the warranty phase comes to an end. For more information, see Chapter 44 in the 2023 *ASHRAE Handbook—HVAC Applications* or ASHRAE *Guideline* 1.1.

Building commissioning contributes to successful sustainable HVAC design by incorporating the system training requirements necessary for building management staff to efficiently take over ownership, operation, and maintenance of the HVAC systems over the installation's useful service life.

Building system commissioning is often contracted directly by an owner and is required by many standards to achieve peak building system performance. Review in the design phase of a project should consider both commissioning and air and water balancing, which should continue through the construction and warranty phases. For additional information, refer to ASHRAE *Guideline* 0.

With building certification programs (e.g., ASHRAE's bEQ, LEED®), commissioning is a prerequisite because of the importance of ensuring that high-performance energy and environmental designs are long-term successes.

REFERENCES

ASHRAE members can access *ASHRAE Journal* articles and ASHRAE research project final reports at technologyportal.ashrae.org. Articles and reports are also available for purchase by nonmembers in the online ASHRAE Bookstore at www.ashrae.org/bookstore.

ASHRAE. 2019. The commissioning process. *Guideline* 0-2019.

ASHRAE. 2007. HVAC&R technical requirements for the commissioning process. *Guideline* 1.1-2007.

ASHRAE. 2019. Guideline for the risk management of public health and safety in buildings. *Guideline* 29-2019.

ASHRAE. 2022. Safety standard for refrigeration systems. ANSI/ASHRAE *Standard* 15-2022.

ASHRAE. 2023. Thermal environmental conditions for human occupancy. ANSI/ASHRAE *Standard* 55-2023.

ASHRAE. 2022. Ventilation for acceptable indoor air quality. ANSI/ASHRAE *Standard* 62.1-2022.

ASHRAE. 2022. Energy standard for buildings except low-rise residential buildings. ANSI/ASHRAE *Standard* 90.1-2022.

ASHRAE. 2024. Energy efficiency in existing building. ANSI/ASHRAE *Standard* 100-2024.

ASHRAE. 2020. BACnet®—A data communication protocol for building automation and control networks. ANSI/ASHRAE *Standard* 135-2020.

ASHRAE. 2021. Climate data for building design standards. ANSI/ASHRAE *Standard* 169-2021.

ASHRAE. 2018. Standard practice for inspection and maintenance of commercial building. ANSI/ASHRAE/ACCA *Standard* 180-2018.

ASHRAE. 2023. Standard for the design of high-performance green buildings except low-rise residential buildings. ANSI/ASHRAE/ICC/USGBC/IES *Standard* 189.1-2023.

ASHRAE. 2018. Commissioning process for buildings and systems. ANSI/ASHRAE *Standard* 202-2018.

BIBLIOGRAPHY

ASHRAE. 2022. Ventilation and acceptable indoor air quality in residential buildings. ANSI/ASHRAE *Standard* 62.2-2022.

ASHRAE. 2018. Energy efficient design of low-rise residential buildings. ANSI/ASHRAE *Standard* 90.2-2018.